

Kenneth Hung

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EDUCATION

University of California, Berkeley

Ph.D. in Mathematics; GPA: 3.9/4.0

Berkeley, CA

Aug. 2014 – May 2019

California Institute of Technology

B.S. with Honors in Mathematics and Computer Science (minor); GPA: 4.0/4.0

Pasadena, CA

Sept. 2010 – May 2014

WORK EXPERIENCE

Meta Platforms Inc.

Research Scientist, Central Applied Science; Research Scientist, Ads Online Experimentation

San Francisco, CA

Jul. 2019 – Present

- **Meta-analysis of experimental data:** Improved experimentation efficiency and quality through empirical Bayesian methods
- **Causal inference:** Semiparametric-efficient estimation in experiments, treatment effect estimation in experiments with spillover, treatment effect adjustment for imperfect experiment

Citadel LLC

Quantitative Researcher Intern

Chicago, IL

May 2017 – Aug. 2017

- **Market making team:** Two projects on high frequency trading stock price predictive models
- **Model selection:** Investigated new high-dimensional feature selection in linear models for best model and best model path

SELECTED PUBLICATION AND PREPRINTS

Empirical Bayes selection for value maximization

Dominic Coey and Kenneth Hung, ACM Knowledge Discovery & Data Mining

2025

- **Regret bound:** Proof of a regret bound when solving a choose- m -out-of- n -items problem using an empirical Bayes approach
- **Semi-synthetic simulations:** Simulation based on publicly available datasets to illustrate the regret in a parametric case, achieving the proved regret bound under correct specification

Statistical methods for replicability assessment

Kenneth Hung and William Fithian, Annals of Applied Statistics

2020

- **Meta-analysis:** Analyzed dataset from experimental psychology replications to quantitatively answer previously vague questions about replicability in the scientific domain
- **Multiple testing and post-selection inference:** Developed new tests and new metrics for replicability analysis
- **Simulations and recommendations:** Simulations and data visualizations in support of better future scientific practices

Rank verification for exponential families

Kenneth Hung and William Fithian, Annals of Statistics

2019

- **Multiple comparison with sample best:** Devised a more powerful approach to this classical problem that handles sparse large parameters without sacrificing power in the dense case
- **Simulations:** Demonstrated gains in power using Matlab, Python and R

SKILLS

Languages: C/C++, Mathematica, Matlab, Python, R

Technologies: git, L^AT_EX

HONORS AND AWARDS

Scott Russell Johnson Undergraduate Prize, California Institute of Technology

Awarded to the best graduating mathematics major

2014

International Mathematical Olympiad

Represented Hong Kong; Bronze and Silver

2009, 2010